Model answer to Q1(a): Computed Example

Consider the satellite communications link illustrated in Figure 1.1. The satellite is in a circular geosynchronous orbit in the plane of the Earth's equator and has an altitude of 35,786 km above sea level. Both the uplink and downlink earth stations are both located on the equator, at sea level. For simplicity, ignore the curvature of the earth.

i) Write down the well-known Friis range equation for a point-to-point communications link.

$$P_{RX} = P_{TX} \cdot \frac{G_{TX} \cdot G_{RX} \cdot \lambda_o^2}{(4\pi R)^2} \quad \textit{Friis' Range Equation}$$

[3]

ii) From 1(a)(i), deduce the expression for the free space path loss term shown in Figure 1.1 and use this to calculate the uplink and downlink ranges.

$$Free\ space\ path\ loss = -10log_{10}\left(\frac{\lambda_o}{4\pi R}\right)^2$$

For 6 GHz uplink: $\lambda_o = 50$ mm and range $R = 39{,}789$ km
For 4 GHz downlink: $\lambda_o = 75$ mm and range $R = 37{,}657$ km

[5]

iii) What is the separation distance between the uplink and downlink earth stations? *Hint: Use simple trigonometry.*

Separation distance is: $\sqrt{(39{,}789\text{ km})^2 - (35{,}786\text{ km})^2} + \sqrt{(37{,}657\text{ km})^2 - (35{,}786\text{ km})^2} = 29{,}115$ km

[4]

iv) If the uplink and downlink earth station antennas are now pointed directly towards each other and the downlink earth station is retuned to operate at 6 GHz, calculate its LNA gain to achieve the originally specified output signal power.

Free space path loss = 197.3 dB

To achieve no end-to end gain, the LNA gain must now have a signal power gain of:

$$0 - 50 - 60 + 197.3 - 42 - 40 = +5.3\text{ dB}$$

[4]

Model answer to Q1(b): Textbook Derivation

Consider a signal generator connected to a load impedance, via a transmission line having a power loss L_L, all being matched to the same reference impedance. Assuming the signal generator, transmission line and load impedance are all in thermodynamic equilibrium and at the physical ambient temperature T_o. The input signal power S_i and input noise power N_i to the transmission line can be combined to give an input signal-to-noise power ratio $SNR = S_i/N_i$.

i) Give the mathematical relationship between equivalent noise temperature and noise factor.

$$\text{Equivalent Noise Temperarture}, T_{RX} = T_o(F_{RX} - 1)$$

[2]

ii) Give an expression for the output noise power.

Answers

$No = G_{RX}F_{RX}N_i \quad when \quad N_i = kT_oB$

[3]

iii) If the transmission line has a loss of 3 dB, what is its equivalent noise temperature in relation to T_o?

$G = L_L = ½$ and so $F = 1/L_L = 2$ and so $T = 2T_o$

[4]

iv) Determine the output signal power S_o and output noise power N_o from the transmission line and give the output signal-to-noise power ratio $SNR = S_o/N_o$. What is the mathematical relationship between the output and input SNR?

$S_o = S_i/2$ and $N_o = G\,F\,Ni = Ni$ and so $S_o/N_o = (S_i/N_i)/2$. Therefore, $(S_i/N_i)/(S_o/N_o)=F$

[4]

Model answer to Q1(c): Computed Example

Consider the downlink earth station in Figure 1.1.

i) Write down the well-known Friss' formula for the noise factor of cascaded stages.

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \frac{F_4 - 1}{G_1 G_2 G_3} + \ldots$$

[3]

ii) Using the relationship in 1(b)(i) with that in 1(c)(i), derive the equivalent noise temperature for two cascaded stages.

$$T = T_1 + T_2/G_1$$

[2]

iii) If the equivalent noise temperatures for the antenna, LNA and receiver are 38 K, 120 K and 4,000 K, respectively, calculate the system equivalent noise temperature.

$$T_{system} = T_{antenna} + T_{LNA} + \frac{T_{receiver}}{G_{LNA}} = 158.04 \text{ K}$$

[4]

iv) With Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K and a receiver bandwidth of 27 MHz, calculate the noise power at the output of the receiver.

$$N_o = G_{LNA}G_{RECEIVER}\left(kT_{system}B\right) = 10^9 \times 58.9 \text{ fW} = 58.9 \text{ µW} = -42.3 \text{ dBW}$$

[4]

v) With the satellite having an EIRP of +34 dBW, calculate the signal power at the output of the receiver.

$$S_o = EIRP - Signalloss + G_{ANTENNA} + G_{LNA} + G_{RECEIVER} = -30 \text{ dBW} = 1 \text{ mW}$$

[4]

vi) Calculate the SNR at the output of the receiver.

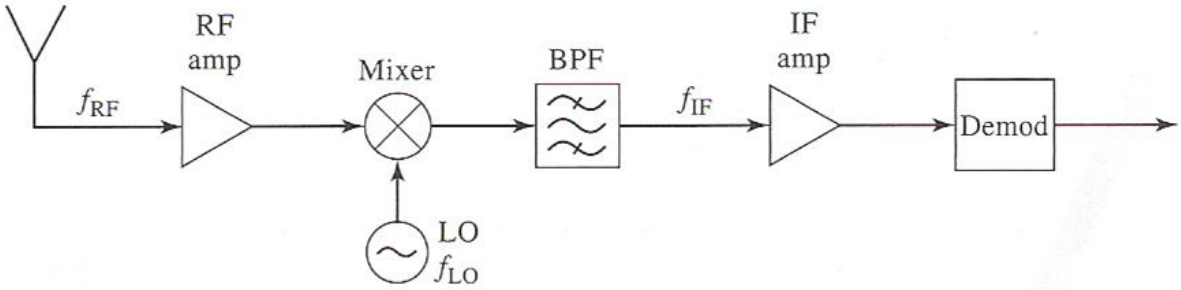
$$\frac{S_o}{N_o} = 12.3 \text{ dB}$$

[3]

Answers

Model answer to Q2(a): Bookwork

Draw the block diagram for a single-conversion superhet receiver and, stating its full name, explain the origin of this type of architecture.



Superhet is an abbreviation for Super-sonic Heterodyne (or Superheterodyne for short) and refers to the fact that the IF frequency is much higher than audio frequencies.

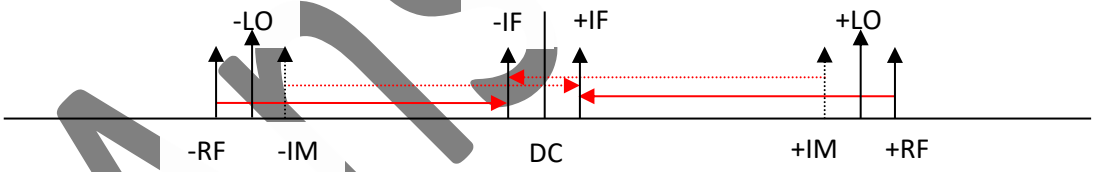
[3]

Model answer to Q2(b): Extended Discussions

(i) With the help of a spectral illustration, briefly explain why an image rejection filter is needed and where this should be located within the architecture. Also, briefly explain why this filter represents a problem and list 3 possible solutions that can be adopted to alleviate or avoid this problem.

Image rejection filter should be inserted in the RF path, before the mixer.

Without this filter, with reference to the spectral illustration should below, the +RF that is translated down into the +IF will be combined with any negative spectral image (-IM) that is translated up to the +IF.



This filter represents a problem because if the local intermediate frequency (IF) is too low then a very expensive high-order filter is needed in order to sufficiently reject the close-to-RF image frequency.

Three possible ways to alleviate or avoid the problem include: (1) using a double-superheterodyne architecture, which will allow a reduction in the filter order by having a higher first IF; (2) use a zero-IF architecture, which does not need an IR filter; and (3) employ and image rejection mixer, which does not need an IR filter.

[4]

(ii) For selecting bands with higher levels of adjacent-band rejection, what simple method is usually can be adopted.

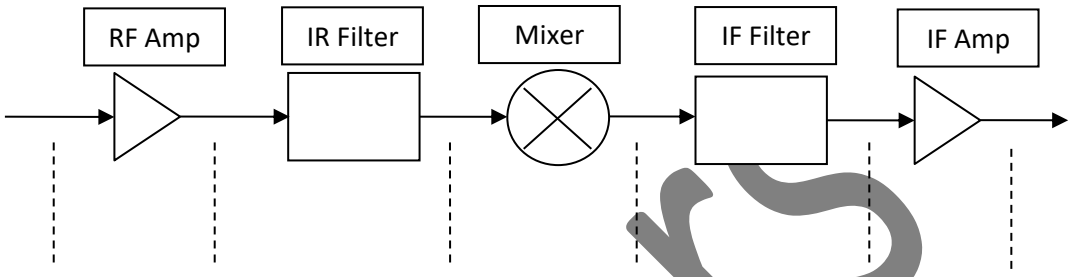
Selectivity is improved by having much cheaper high order band-pass filters located in the IF.

[2]

Answers

Model answer to Q2(c): Calculated Application of Theory

The receiver has an input power level of -50 dBm. With the simplest of architecture design for the type in 2(a), up to the demodulator, it can be assumed that there is only one filter and one amplifier at any given frequency. All amplifiers have a gain of 30 dB, IP3 of 40 dBm and noise figure of 1 dB; all filters having a pass band insertion loss of 2 dB and the mixer has a conversion loss of 6 dB and IP3 of 30 dBm. Ignoring noise contributions from the local oscillator, calculate the following at the output of each sub-system block (where appropriate) and state the main equations used



G [dB]	+30	-2	-6	-2	+30	
IP_3 [dBm]	+40	+100	+30	+100	+40	
F [dB]	+1	+2	+6	+2	+1	

+100 are assumed values and this may change for different students!

Pout [dBm]	-50	-20	-22	-28	-30	0 [2]
IP_3 [dBm]		+40	+38	+27.88	+25.88	+39.89 [3]
IMD_3 [dBc]		120	120	111.76	111.76	79.78 [2]
I_3 [dBm]		-140	-142	-139.76	-141.76	-79.78 [3]
F_{RX} [dB]	(1.259 +0.000585 +0.00472 + 0.00369 +0.00259) = 1.271 = 1.04 dB					[3]

The following main equations should be used:

$$C = P_{OUT}(fo) = G_1 G_2 P_{IN} \quad and \quad IP_3 = (IP_3|_1 \; G_2) \| \, IP_3|_2$$

$$IMD_3 = \frac{C}{I_3} \sim \left(\frac{IP_3}{C}\right)^2 \equiv 2\left(IP_3[dBm] - C[dBm]\right)[dBc]$$

$$-I_3 = \left(IMD_3[dBc] - C[dBm]\right)[dBm]$$

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \frac{F_4 - 1}{G_1 G_2 G_3} + \ldots$$

If the System Temperature is the same as the ambient temperature of To = 290 K (which would be rather unusual) then (kToB) = -129.2 dBm the output noise temperature would be: $No = G_{RX} F_{RX} (k T_o B)$ = -78.16 dBm, giving a signal-to-noise ratio at the output of the receiver So/No = 78.16 dB.

[3]

Answers

Model answer to Q 3(a): Textbook and Calculated Example

(i) For a lossless linear two-port network, state the simple equation that show the relationship between insertion loss and return loss.

$$Insertion\ loss = 10log_{10}|S_{21}|^2$$

$$Return\ Loss = 10log_{10}|S_{11}|^2$$

$$For\ a\ lossless\ linear\ 2-port\ network, 1 \equiv |S_{21}|^2 + |S_{11}|^2$$

[3]

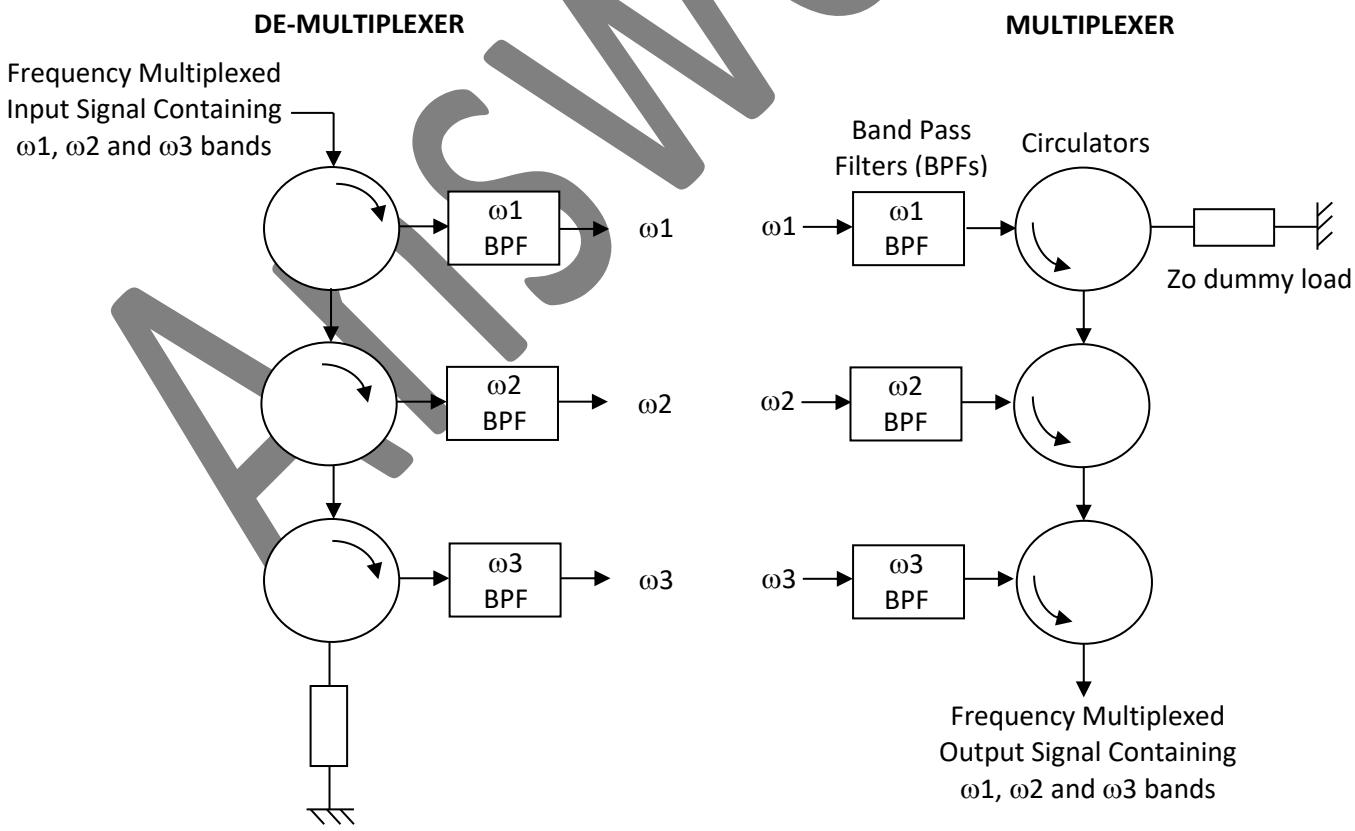
(ii) Using the equation stated in 3(a)(i), calculate the return loss when insertion loss is -1 dB.

$$|S_{21}|^2 = 0.794\ and\ |S_{11}|^2 = 0.206\ and\ Return\ Loss = -6.868\ dB$$

[4]

Model answer to Q 3(b): New Application

Draw a 3-channel frequency de-multiplexer and a 3-channel frequency multiplexer, employing circulators, band-pass filters and dummy loads.

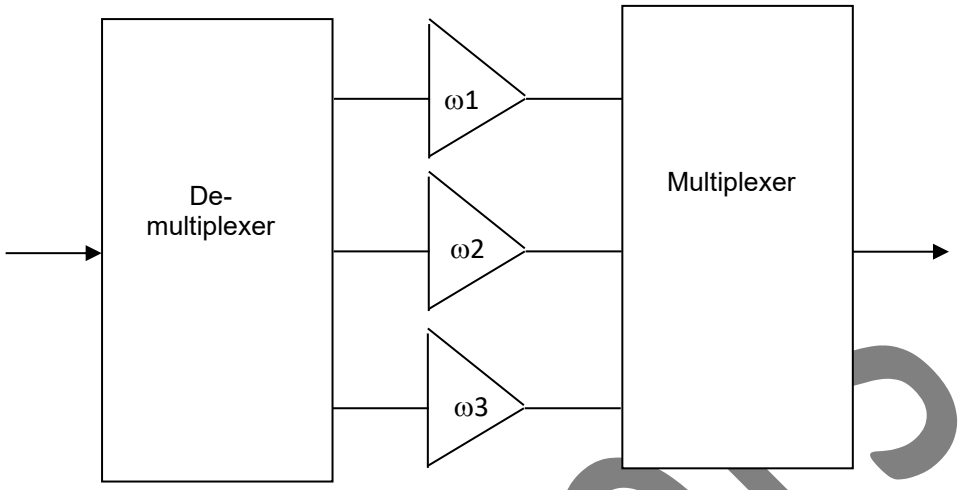


[5]

Answers

Model answer to Q 3(c): New Application

Employing the topologies given in 3(b), draw a unilateral 3-channel amplifier. Assume that for each frequency band the associated filters and amplifiers have the same -1 dB band edge frequencies.



[4]

Model answer to Q 3(d): Calculated Example

Employing the topology from in 3(c), use simple power loss budget analysis to calculate the overall gain, given that the insertion loss of each circulator is - 0.1 dB, the mid-band insertion loss of the filters is 0 dB and the mid-band gain of the amplifiers is 10 dB. State any simplifying assumptions.

(i) Calculate the gain at mid-band, assuming that the transition frequency bandwidth is insignificant.

Total Insertion Loss = -0.1 – 0 +10 – 0 -0.1 -0.1 – 0 -0.1 -0.1 -0 -0.1 = +9.4 dB

Assuming perfectly matched circulators and no inter-stage reflections between the filters and the amplifier

[3]

(ii) Calculate the gain at the -1 dB frequency band edge, assuming that the ideal upper -1 dB band edge frequency of one channel is the same as the ideal lower -1 dB band edge frequency of the adjacent channel.

Total Insertion Loss = -0.1-1 +9 -1 -0.1 -0.1 – 6.868 -0.1 -0.1-0 -0.1 = -0.468 dB

Assuming perfectly matched circulators and no inter-stage reflections between the filters and the amplifier, but taking into account the -6.868 dB return loss at the output of the final filter seen by the circulator.

[4]

(iii) Using the results from 3(d)(i) and (ii), comment on the resulting frequency response over the entire bandwidth of the 3 channels.

The frequency response for one channel amplifier overlaps the other. However, in the region of the overlapping -1 dB band edge, rather than a large power gain there is a power loss. As a result, the roll-off for each channel will be much steeper and so the overall ripple across the entire bandwidth of the 3 channels will be severe.

[2]

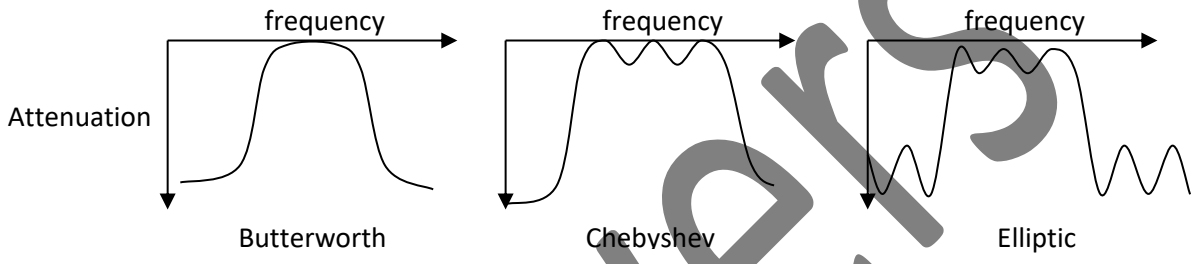
Answers

Model answer to Q 4(a): Bookwork and Discussions in Class

With the use of simple illustrations for the attenuation against frequency curves, describe the differences between Butterworth, Chebyshev and Elliptical-function filters. Also, comment on the group delay characteristics for these filters.

Butterworth filters have a maximally flat frequency response in the passband. Chebyshev filters exhibit ripples in the passband only; the number depends on the filter order. Neither the Butterworth nor Chebyshev filters have ripples in the stopband. Elliptical filters have the sharpest roll-off attenuation characteristics of all three types, with the Chebyshev having sharper roll-off than the Butterworth. The elliptic filter exhibits ripples in the stopband (transmission zeros) as well as ripples in the passband (transmission poles).

Chebyshev and elliptic filters have higher group delays than the Butterworth, particularly near the 3 dB cut-off frequency and should be avoided in applications where pulse distortion is critical



[5]

Model answer to Q 6(b): Computed Example

Given prototype low-pass filter attenuation curves and tables for the corresponding normalised element values (see attached sheets), design an L-C lumped-element band-pass filter that meets the following specifications:

Centre Frequency, *fo*	500 MHz
3 dB Bandwidth, *B*	50 MHz
Attenuation Bandwidth	100 MHz
Pass-Band Ripple (Peak-to-Peak)	0.1 dB
Stop-Band Attenuation	45 dB
Input Impedance, R_{IN}	100 Ω
Output Impedance, R_{OUT}	50 Ω

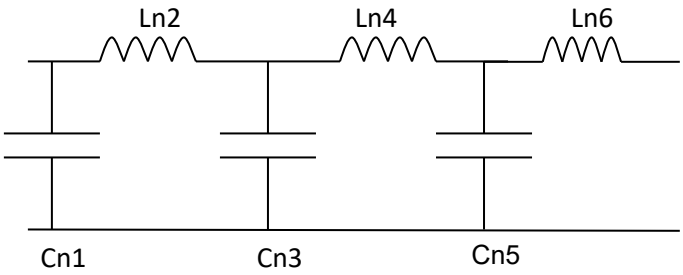
Determine f/fc = B/BW_{STOP} = 2.

From curves: Butterworth with 0.00 dB ripples need >7th order.
Chebyshev with 0.01 dB ripples need 6th order for 46 dB rejection having f/fc = 2.
Chebyshev with 0.10 dB ripples need 6th order for 50 dB rejection having f/fc = 2.
This last option is best, as it has good out-of-band attenuation margin.

For Rin/Rout = Rs/R_L = 2, the normalised prototype low pass filter coefficients are:

Cn1	Ln2	Cn3	Ln4	Cn5	Ln6
0.414	3.068	0.958	3.712	0.979	2.794

Answers



To de-normalize shunt capacitors and inductors:

$$C_p = \frac{Cn}{2\pi f B R_L} \qquad L_P = \frac{R_L B}{2\pi f_c{}^2 Cn}$$

To de-normalize series inductors and capacitors:

$$L_s = \frac{LnR_L}{2\pi B} \qquad C_S = \frac{B}{2\pi f_o{}^2 LnR_L}$$

Cp1 = 26.36 pF	Lp1 = 3.84 nH
Cs2 = 0.21 pF	Ls2 = 488.29 nH
Cp3 = 61.00 pF	Lp3 = 1.66 nH
Cs4 = 0.71 pF	Ls4 = 590.78 nH
Cp5 = 62.33 pF	Lp5 = 1.63 nH
Cs6 = 0.23 pF	Ls6 = 444.68 nH

...vers



[20]